Allah's Plan is greater than your dreams.

Database connection.

Exception Handling.

2 types.

- Runtime
  - divided by zero.
  - (optional)
  - for good programmers.
- I/O exception
  - (must) handle

```
[ ≡≡≡ try { ≡≡≡ } catch ( ) ]
```

socket
↳ codeNo. in service.

PL SQL →

connect system/123@myd6 ④ 2000

XE

① user ② pass ③ dbname

⑤ IP address of host machine.

essential infos to connect to any db from

anywhere.

"test"

concat → || , +

|| → SQL

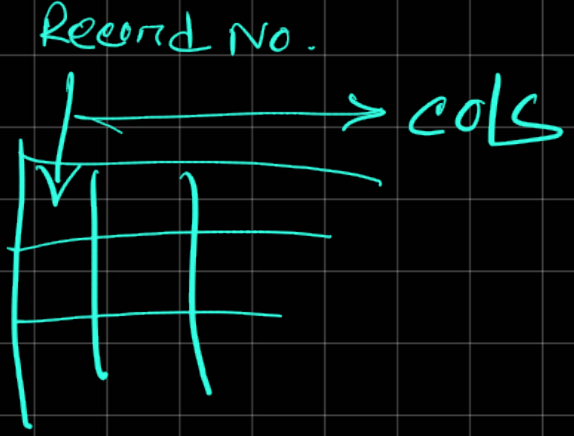
+ → Java

executeUpdate("") ;

↑ expects string

if I want to insert record,

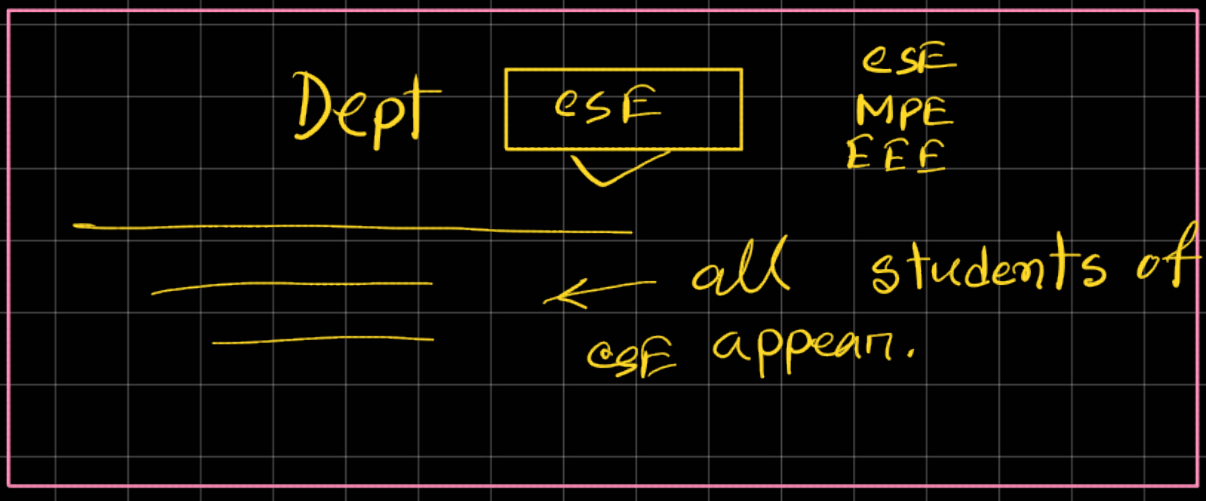
insert int x values ('a')

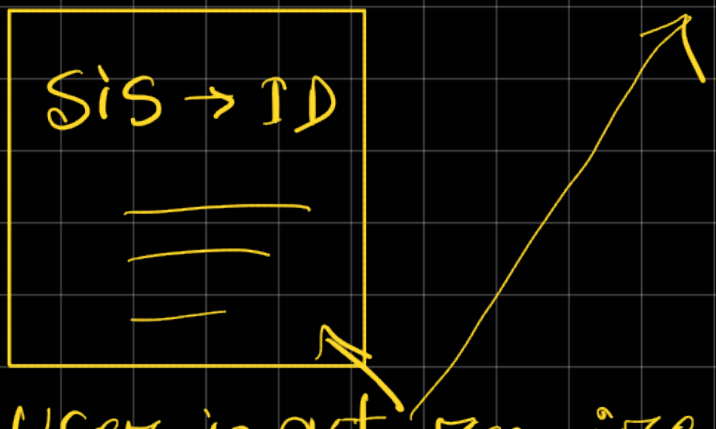


print using grouping.

SQL injection → Assignment.

Rset.



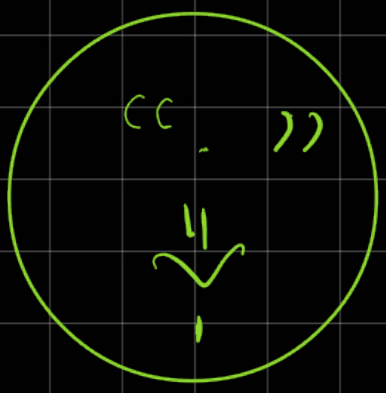


Select ID, Name, CGPA
from students
where dept = 'CSE' → hard coded
we don't want

vdept → string will give which dept
I want

Select ID, Name, CGPA
from students
where dept = ' " + VDept + " ' .

= 'CSE'



login a system:

Users	
UserID	Pwd
u1	hhm2
u2	xyz

UserInput

select *
from users
where userid = u1
& pwd = pwd

if users give vdept = X' OR 'B' = 'B
if pwd even without knowing I can

where dept = 'X' OR 'B' = 'B'

modified where clause → true
enter the database → sql injection

what is basic principle on which

the sql injection works → concat strings
↳ lots of space + string.

a'; → delete or update → Any DML will work.

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in C, Functions

```
int getResult (int x) {
    return result;
}
```

(FPM) user input → formal param

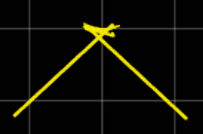
```
R = getResult(105);
```

↓ argument / actual parameter.

in PLpgSQL → its ok called IN p.m.

in DBMS another one OUT ||.

For result
point = ?
sum credit.
return point by credit.



④ A → CSE 4410 — 3 credit
B → " 4400 → 2 "

③ D

12+6 = point.

Tot cred = 5

$\frac{\text{point}}{\text{cred}}$ = CGPA.

Related info + Result needed:

Total number of courses & credits etc needed. → OUT (param)

↳ A Func can return multiple out parameter ✓

1 single value.